FlexTail Sensor Documentation

Advanced Wearable Technology for Biomechanical Analysis

MinkTec GmbH

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1 FlexTail Sensor System

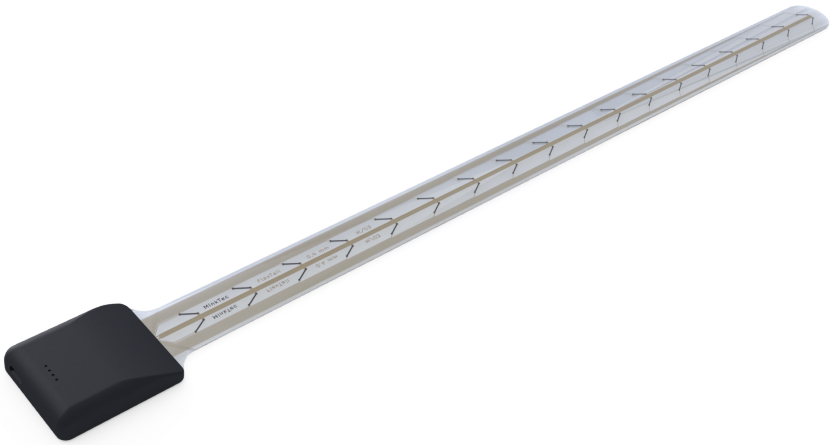

Figure 1: FlexTail Sensor

The FlexTail (shown in Figure 1) is an advanced wearable sensing device designed for real-time biomechanical analysis. It represents a breakthrough in wearable flexible sensor technology, capable of measuring complex deformation patterns across multiple points along its length and over a long period of time. It is usually attached to the body with a shirt or plaster in which the sensor can slide.

1.1 Overview

The FlexTail combines two complementary sensing modalities to provide comprehensive biomechanical data:

1. **Flexible Sensing Strip**: Equipped with 18 precision strain gauge pairs distributed along its length, detecting local bending and torsional deformation
2. **Integrated IMU**: A rigid case containing high-precision accelerometer and gyroscope sensors for orientation and motion tracking

This dual-sensor architecture enables the FlexTail to capture complex mechanical behaviors including curvature profiles, torsional twisting, and dynamic movement patterns, making it ideal for biomechanical research, ergonomic assessment, and posture monitoring applications.

1.2 Technical Specifications

1.2.1 Nomenclature and Orientation

When talking about the FlexTail, we refer to it as “FlexTail” or “sensor” interchangeably throughout this document. Left and right are always used as in Figure 2. We imagine the sensor lying in front

of us with the case pointing towards us and the sensor strip pointing away from us. A pair of strain gauges used to measure strain is referred to as a sensor pair. The first measuring point or coordinate is always the one closest to the case. The area where the angles are measured is called the measuring area (Figure 3).

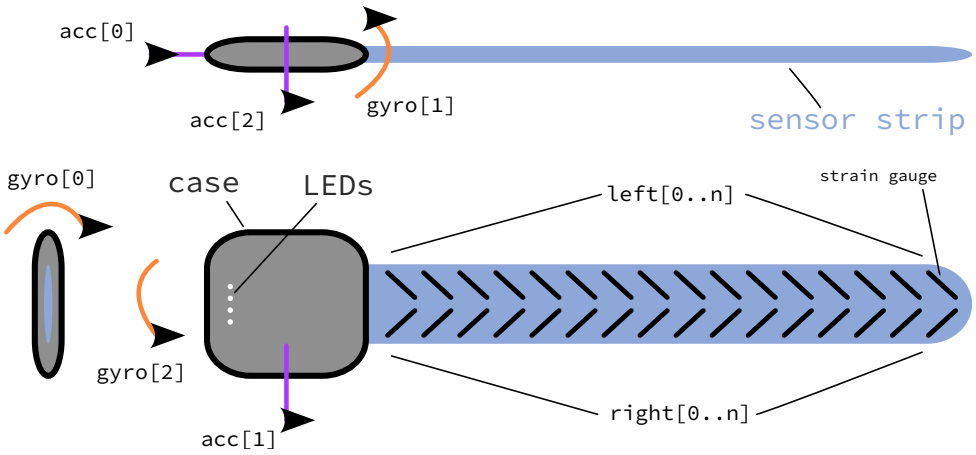

Figure 2: Sensor Overview

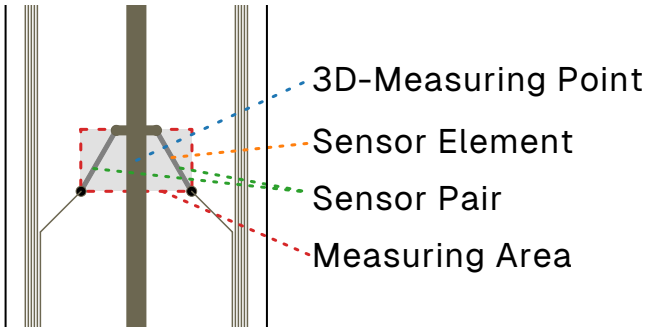

Figure 3: Sensor Closeup

1.2.2 Physical Characteristics

The FlexTail is available in two configurations to accommodate different application requirements:

- **36 cm (Short)**: Sensor elements spaced 20 mm apart, providing detailed measurements over a shorter range
- **45 cm (Long)**: Sensor elements spaced 25 mm apart, offering extended coverage area

Both variants maintain the same 18 sensor pair configuration, with the spacing determining the spatial resolution of measurements along the strip. We manufacture the sensor strip in different lengths, widths and thicknesses. Contact us for custom designs at info@minktec.com.

1.2.3 Sensing Capabilities

1.2.3.1 Strain Gauge Array The flexible sensing strip incorporates 18 strain gauge pairs arranged in a linear configuration. Each pair consists of:

- **Left sensor elements**: Detect deformation on the left side of the strip
- **Right sensor elements**: Detect deformation on the right side

This dual-sided configuration enables:

- **Bending detection**: Forward/backward flexion measurement through differential strain analysis
- **Torsional measurement**: Rotational twisting detection via comparative analysis between sensor pairs
- **Directional sensitivity**: Positive values indicate forward bending, negative values indicate backward bending.

1.2.3.2 Inertial Measurement Unit The integrated IMU provides complementary orientation and motion data:

- **3-axis accelerometer**: Measures linear acceleration with 1g ≈ 2000 units resolution
- **3-axis gyroscope**: Captures angular velocity around each axis for rotation detection
- **Orientation calculation**: Derives pitch and roll angles for spatial positioning

1.2.4 Coordinate System and Orientation

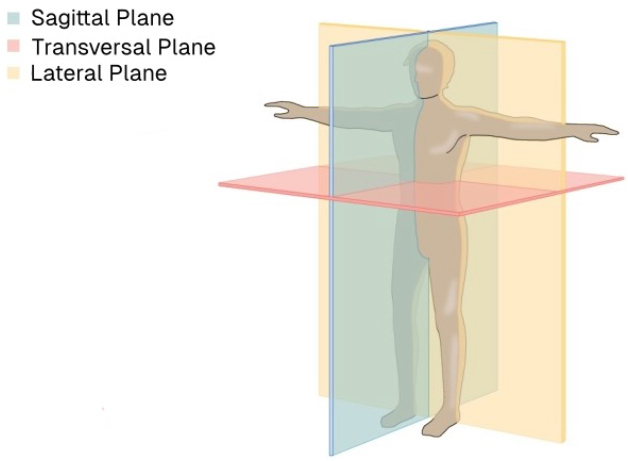

Figure 4: Anatomical Reference Planes

Throughout this document and the software, you will encounter some anatomical terms. The most important are “sagittal” and “lateral,” which refer to the planes shown in Figure 4. The sagittal plane divides the body into left and right halves. The lateral plane runs parallel to the back.

For anatomical reference, the FlexTail system uses orientation planes that match human anatomy. All mentions of sagittal or lateral directions follow these conventions. The system provides calculated coordinates in millimeters by default. The coordinate starts somewhat inside the case as depicted in Figure 5. The exact location is the first sensor pair where the strain gauges meet in the center of the sensor strip and from this point -20 mm for the short sensor and -25 mm for the long sensor along the Z-axis. The coordinates (3D-Measuring Point) are placed in the middle of rectangle drawn in Figure 3.

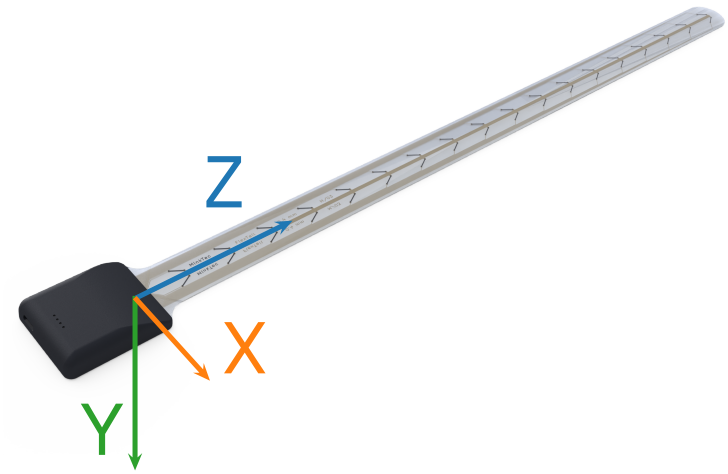

Figure 5: FlexTail Coordinate System

The FlexTail employs a right-handed Cartesian coordinate system (Figure 6) specifically designed for biomechanical analysis:

- **X-axis**: Left to right (transverse direction)
- **Y-axis**: Front to back (sagittal direction)
- **Z-axis**: Bottom to top (longitudinal or vertical direction)

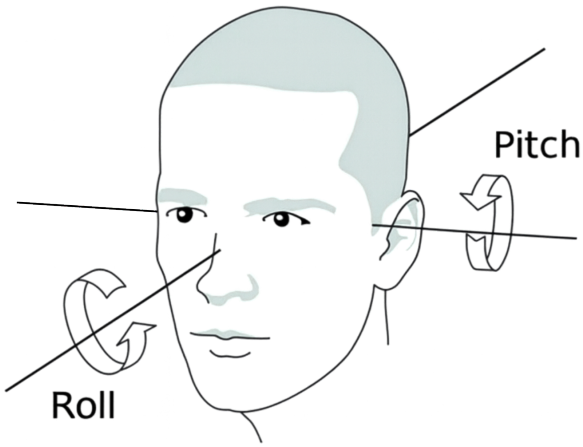

Figure 6: Sensor Coordinate System

1.2.4.1 Inertial Measurement Unit Orientation In addition to deformation sensing, the FlexTail includes a built-in inertial measurement unit (IMU) to track its own orientation in space. These values are essential for aligning the sensor in 3D space, detecting large motion events, and augmenting bending data with inertial measurements.

The accelerometer data is used to estimate spatial orientation through two key parameters:

- **Pitch**: Tilt around the X-axis; positive values indicate a forward tilt
- **Roll**: Tilt around the Y-axis; positive values indicate a rightward tilt

This orientation system enables precise spatial referencing and allows for accurate biomechanical analysis regardless of the sensor's initial positioning on the body.

1.2.5 Battery

The sensor battery lasts approximately 30 hours when recording at a frequency of 5 Hz. At higher frequencies, battery life decreases; for example, it lasts about 24 hours at 25 Hz. The battery can be charged via USB-C, and a full charge takes around 90 minutes. The FlexTail is PD (Power Delivery) compliant, meaning it can be charged with a standard phone or laptop USB-C charger.

Battery status is indicated by four LEDs on the back of the device. A short tap on the power button activates the battery display if the sensor is powered on. The current charge level is also shown when the sensor is turned on.

1.2.6 Shirt and Sensor

The Sensor comes with specially designed shirts that feature a longitudinal pocket to put the sensor in. To mount the sensor just slide it in and close the pressure button. The shirts are made out of 95~% Cotton and 5~% Elastan. The sensors need to be put in with the flat side facing the user's back. The shirts come in extra sizes. The sizes available are: XS, XS+, S, S+, M, M+, L, L+, L++, XL, XL+, XL++. Each "+" indicates an additional 4 cm of width around the waist. These size options help ensure better measurement accuracy. For optimal results, the shirt should fit snugly, as a tighter fit leads to more precise measurements. The shirts are designed for comfort during everyday wear. If maximum precision is required, we recommend securing the sensor directly to the body—such as with a hollow plaster or other mounting methods—to prevent any movement. Only mounting methods that allow the sensor strip, but not the case, to slide in the Z-direction of the FlexTail should be used.

1.2.7 Buffering

If you lose connection to your FlexTail during recording, the device automatically saves data to its integrated flash memory. Once the connection is restored, the buffered data is transmitted to your recording device. Note that this transfer is not instantaneous—on mobile devices, transmission rates can reach up to 80 Hz. For example, if you record at 25 Hz, sending 1 second of buffered data takes approximately 0.5 seconds.

1.3 Software Ecosystem

The FlexTail is supported by a comprehensive software system providing:

- **Data processing**: Efficient loading and manipulation of sensor data
- **Analysis tools**: Pre-built algorithms for common biomechanical calculations
- **Visualization**: Real-time and post-processing graphical representations
- **Export capabilities**: Multiple output formats for further analysis

1.4 FlexLab Studio

FlexLab Studio is a cross-platform tool, able to record, view, analyze and export data. It is available on the following platforms:

Platform	How to get it	Comment
Android	Play Store	
Windows	Microsoft Store	
Web	studio.rectify.de	Chrome recommended [1]
iOS	TestFlight	Contact us (Apple ID required)
macOS	TestFlight	Contact us (Apple ID required)
Linux	Contact info@minktec.com	Beta Version, Bluetooth is unstable

1.5 flexlib

It is a Python library for processing and analyzing sensor data from the FlexTail. The library provides tools for loading, processing, and evaluating biomechanical measurements.

Core Functionality:

- **Data Processing**: Load and process sensor measurements from various file formats (RSF v1/v2, CSV)
- **Biomechanical Analysis**: Calculate spine angles, posture metrics, and movement parameters
- **Signal Processing**: Apply filters for motion and bending detection.
- **Data Evaluation**: Extract metrics from measurement data for analysis

The documentation and wheels can be downloaded from our Google Drive folder

1.6 fiffi_unleashed

It is a Python BLE (Bluetooth Low Energy) client library for connecting to and controlling FlexTail devices in real-time. The library enables direct device communication for live data streaming and sensor control.

Core Functionality:

[1] Not all browsers support Bluetooth Low Energy. See: caniuse.com/web-bluetooth for a full list.

- **Device Communication**: Connect to FlexTail devices via Bluetooth Low Energy or serial connection
- **Real-time Streaming**: Receive live measurement data with customizable frequency and processing
- **Sensor Control**: Calibrate sensors, start/stop measurements, and configure device settings
- **Event Handling**: Listen to connection events, measurements, and device status updates

The library is currently in active development (beta release). The interface may change in future releases. We recommend implementing proper separation between the flexlib and fiffi_unleashed APIs in your code. Planned features include automatic hardware version detection, improved session handling with automatic reconnection, and support for retrieving buffered data from the device's flash memory.

The documentation and wheels can be downloaded from our Google Drive Folder

1.7 rectify

The rectify-App is our consumer-facing Digital Back Coach. In conjunction with the shirt it helps users to improve their posture habits and prevent back pain through awareness, gamification and training. It is available for Android and iOS.

- **Real-time posture feedback:** Your spine movement is continuously monitored, and the app triggers vibration alerts when you sit in a poor posture for too long
- **Customized training plans:** The app creates individualized exercises based on your posture patterns and mobility, rather than generic routines
- **Motivational tools:** Set daily goals, participate in posture challenges or leaderboards—especially fun in groups—to track progress and stay engaged

1.8 Custom Software

We offer the development of custom software for specific use cases.
Contact us at info@minktec.com.

1.9 Data Output and Integration

1.9.1 File Format Support

The FlexTail system supports multiple data formats for seamless integration:

- **CSV**: Standard comma-separated values for broad compatibility
- **RSF v1**: Custom file format used for efficient storage.
- **RSF v2**: Custom file format used for efficient reading and writing.

The RSF file formats support annotation. When you record a measurement you will see a Tag button. That places a marker at the current time, to mark important events that occur during your measurement.

1.9.2 Precomputed Metrics

The data from our software can be exported to other formats that can be processed by spreadsheet software (Excel, Google Sheets, etc.).

Pre-computed biomechanical parameters include:

- Lumbar angle, twist, lateral and sagittal flexion
- Approximate calculations for rapid assessment
- Acceleration and gyroscope magnitude measurements
- Timestamp and annotation support for event marking

Name	Description	Positive Direction
Lateral	Angle between the case and the tip in the lateral plane	Right
Sagittal	Angle between the case and the tip in the sagittal plane	Front
Lateral Approx	Approximated angle between the case and the tip in the lateral plane [2]	Right
Sagittal Approx	Approximated angle between the case and the tip in the sagittal plane	Front
Pitch	Angle of the case in the Sagittal plane	Front
Roll	Angle of the case in the Lateral plane	Right
Movement	Rotational movement (calculated from the gyro)	—

[2] The approximation assumes that the sensor strip cannot deform in the sagittal direction. The calculation is approximately 10 times faster and has an average difference of 4% from the unapproximated value under typical bending conditions.

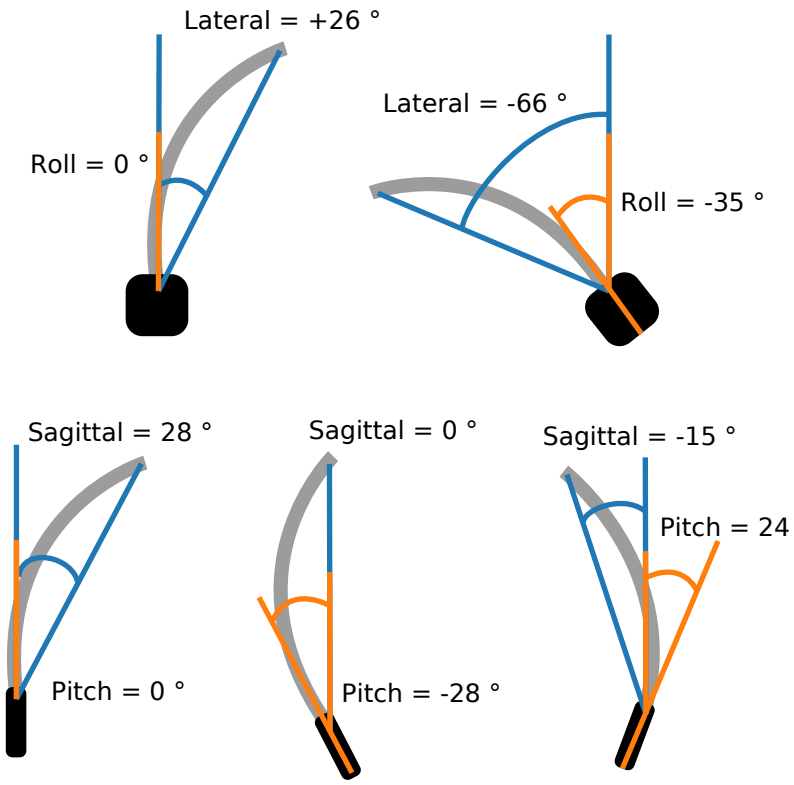

Figure 7: Visualization of precomputed orientation metrics. The sagittal and the lateral angle (blue) is defined between the sensor tip and a vertical reference, with positive values indicating forward flexion and right flexion respectively. Roll and pitch angles (orange) describe the sagittal and lateral deviations of the case only.

2 FlexTail Guide

2.1 Quick Start

1. **Put on the shirt** so it fits snugly for accurate measurements.
2. **Power on the FlexTail sensor.** Calibrate it if needed—see Section 2.4 for instructions.
3. **Insert the sensor** into the shirt pocket from below and fasten the pressure button.
4. **Check placement:** Make sure the sensor case is centered and positioned above the sacrum (Figure 8).
5. **Open FlexLab Studio.** Press the Bluetooth button in the app and follow the connection instructions.
6. **Start recording** your measurement in the “Record” tab.

2.2 Data Privacy

The rectify-App automatically collects some data. For detailed information about how we handle your data, please refer to our privacy policy at rectify.de.

When using the FlexTail without an account, all measurement data remains stored locally on your device and is not transmitted to our servers.

2.3 Placement

The sensor casing should be placed at the sacrum or on belt height for low waist trousers (Figure 8).

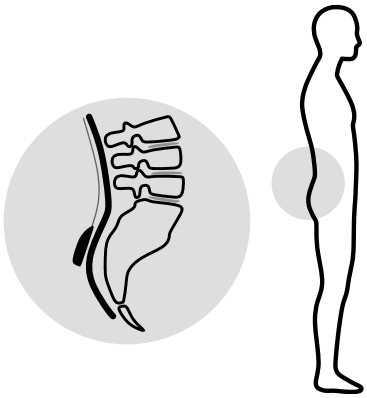

Figure 8: Correct placement of the sensor case

Ensure that the sensor is centered horizontally and does not move too much. Wearing a belt improves the measurement quality since it can be used to clamp the case in place.

2.4 Calibration

The sensor strip loses its zeroing over time. Similar to a kitchen scale where you have to press the "TARE" button, the FlexTail needs to know its own zero-position too. The sensor needs to be calibrated, especially when the wearer sweats (warmth and humidity). Lay the sensor on a flat surface. If it displays a bent shape in the software it needs to be calibrated. Press the calibration button in the software or press the power button of the FlexTail three times. If the LEDs start blinking it's calibrating. It's done when the LEDs have stopped blinking.

How often the sensor needs to be calibrated depends heavily on the use case. So we cannot give a concrete recommendation. If you do not measure people but some other kind of surface the sensor might never need to be calibrated.

3 FAQ

- **The FlexTail won't connect**: Try restarting the software, turning Bluetooth off and on, restarting your device. Check if someone else is connected with your FlexTail.
- **The FlexTail appears to be bent even though it lies on a flat surface:** You need to calibrate your sensor. Place the sensor on a flat surface and press the calibration button in the software or press the power button on the device three times.

- **Are the angles/measurements relative to the wearer's own posture**: No. If the software reports for example a lordosis of 0° this means that the sensor is perfectly straight. If you take a measurement you could press the Tag button when the person sits or stands upright to mark the person's neutral posture and calculate the measurements relative to that neutral posture.
- **Is the FlexTail waterproof**: No. But it's spray water-resistant.
- **Can the sensor detect the user's direction (north, south)**: No, the case only uses a gyroscope and accelerometer. It does not have a magnetometer.
- **Can I use the recorded data with other software?** Yes, you can export CSV files with precalculated values from FlexLab Studio or export your data to the flexlib python library and transform the data in any way you like.
- **Do I need an Account to take Measurements?** No.